

A Single LED-based Indoor Visible Light Positioning System – Recent Trends and the Impact of Ambient Light on Positioning Accuracy

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Abstract—Visible light positioning (VLP) is an emerging localization technique for indoor applications, which utilizes the visible light spectrum to send signals between the optical transmitter (Tx) and the receiver (Rx). In this paper, the recent trend towards the implementation of a single light emitting diode (LED)-based VLP techniques is discussed including the effect of ambient light on the positioning performance. The work establishes a single LED-based VLP scenario, in which the light beacon signal is transmitted from the LED Tx to estimate the location of the user in an indoor environment. The user's optical Rx is equipped with four photodiodes to receive the free-space light signal in parallel via spatial diversity. The received signal strength indicator-based trilateration scheme is implemented to determine the position of the user. Through simulation, the results demonstrate the 90th percentile accuracy of 7, 11, and 13 cm for a dark room, a single ambient source, and two ambient sources, respectively. Hence, approximately a 46% reduction in the positioning error is observed for the case of a dark room without the ambient light-inducing noise.

Keywords—Indoor localization, light emitting diode (LED), optical wireless communications (OWC), received signal strength indicator (RSSI), trilateration, visible light positioning (VLP)

I. INTRODUCTION

Light-based positioning systems date back to ancient times when our ancestors navigated on the Earth by following the guiding stars in the sky. In the present day, the growing number of internet-of-things-enabled devices and location-based services triggered the need for reliable wireless localization systems. Global positioning system (GPS) is a well-established localization technique, which is widely utilized in real-time navigation, robotics, asset management, and other tracking applications due to its ease of access. GPS operates by maintaining a radio link between users and the satellites orbiting around the Earth. The unavailability of a line-of-sight (LOS) path can result in the interruption of this positioning service. For instance, satellite signals suffer from significant power losses in indoor environments mainly due to channel attenuation and

multipath interference [1]. Hence, it necessitates the need for suitable and reliable indoor positioning systems (IPSs).

Different IPS techniques are proposed to tackle these challenges. For instance, Wi-Fi, Zigbee, Bluetooth, Ultra-Wideband (UWB), radio frequency identification (RFID), ultrasound, and infrared-based light detection and ranging (LIDAR) [2]. These systems are limited to certain operating conditions and suffer from a lack of positioning accuracy. Wi-Fi-based localization schemes are subject to multipath fading and interference [3], whereas both Zigbee and Bluetooth-assisted systems are susceptible to fluctuations in signal sources, with the latter requiring user association. Moreover, UWB and RFID-based positioning require dedicated equipment, and the ultrasound-assisted IPS is prone to the effect of ambient temperature on the speed of sound; additionally, LIDAR consumes a large amount of energy and is costly.

The increasing demand for reliable and highly accurate IPS has led to the development of indoor optical wireless localization systems. Within this context, visible light communication (VLC) leveraging the existing lighting infrastructure can be utilized for simultaneous illumination, data communication, and localization in indoor environments [1]. VLC offers the distinctive multifold advantages of high bandwidth (400 THz-red to 800 THz-violet), immunity to radio frequency interference, energy efficiency, and cost-effectiveness. The basic requirement of IPS is at least one reference node with a known location in the space, i.e., wireless access points, as well as the user device whose location is to be calculated by the positioning algorithm. In the traditional downlink visible light positioning (VLP) systems, the intensity-modulated light signals containing the location information of LED transmitters (Tx) are broadcasted to the user. The signal parameters associated with the received power are utilized by the user to implement a positioning algorithm to estimate its location within a given space [4]. In recent years, several VLP algorithms have been proposed with analytical and experimental verification, which have demonstrated to estimate a user's

location with a low positioning error (i.e., 0.1 – 0.35 m) as compared to Wi-Fi (1 – 7 m), Bluetooth (2 – 5 m), and other technologies [5], [6]. Hence, VLP can be utilized as a promising solution for indoor localization as part of future networks, which require centimeter-level accuracy.

VLP-based positioning systems can be classified according to the number of light sources used for transmission, i.e., single or multiple LED Tx's [7]. In the case of multiple Tx's, modulated LEDs flickering at high frequencies require an efficient multiplexing technique to avoid intercell interference. Moreover, the limited field of view (FOV) of the receiver (Rx) restricts the use of multiple LEDs in VLP, since LEDs must be within the Rx's FOV for position estimation to be possible [4]. Therefore, single LED-based VLP offers an energy-efficient, robust, and flexible localization solution that overcomes the interruptions experienced in conventional multiple LED-based VLP systems [8]. It is worth mentioning that a single LED Tx, in practice, refers to an LED panel consisting of multiple individual LED units combined into arrays or integrated into high-powered LED light fixtures.

In this paper, the effect of ambient light on the localization accuracy of single LED-based VLP systems is discussed alongside the recent research advances in the field. This research work explores the single-input multiple-output configuration where the optical Rx is comprised of four photodiodes (PD)s arranged symmetrically in a rectangular geometry, which allows for the spatial diversity to receive the transmitted optical signal in parallel. The study establishes a received signal strength indicator (RSSI)-based trilateration scheme, through the MATLAB[®] simulation platform, comprised of a single LED Tx and a user device, equipped with an optical Rx, whose location needs to be estimated in the indoor environment. To the best of the author's knowledge, it is the first study conducted to analyze the effect of ambient light-inducing noise in single LED-based VLP systems. The results have demonstrated that the positioning error is significantly influenced by the lighting condition, as an error reduction of 46% is recorded for the ambient-noise-free region (i.e., dark room) as compared with the well-lit space.

The next sections of the paper are planned as follows: Section II provides the key literature review for the single LED-based VLP systems, Section III outlines the proposed system model and Section IV evaluates the localization performance, demonstrated by simulation results, in terms of positioning accuracy with and without ambient light-inducing noise in the system. Finally, Section V delivers the concluding remarks.

II. RELATED WORK

Researchers have been working on devising suitable strategies for the single LED-based indoor VLP system as it proposes a flexible solution that ranges from ordinary indoor lighted spaces to hallway configurations, in which the user's visual FOV is guaranteed to receive signals from at least one light source. To enhance the positioning accuracy, the VLP Rx can make use of multiple PDs or auxiliary devices, such as an inertial measurement unit (IMU), instead of multiple LED Tx's. In this context, the work done so far can be classified into two major categories based on the type of optical detection devices at the Rx, i.e., image sensors (ISs) (or cameras), and PDs, both of which are susceptible to ambient light-induced noise. The key

works related to these categories are detailed in the following subsections and Table I provides a summary of the downlink single LED-based VLP techniques.

A. IS-based Rx

Several algorithms have been proposed in the literature, which exploit the geometrical features of the single LED Tx and the use of camera-based VLP systems. In [9], a single circular LED with a red point marker at its edge was used at the Tx to estimate the user's orientation, and the circle projection algorithm was employed to determine the user's location. Experimental results demonstrated the average positioning accuracies of 12.79 and 17.03 cm for two different user heights of 1.5 and 2 m, respectively. For the case of a single circular LED, another VLP system utilizing planes intersection-line theory was developed in [10], which used the angle of arrival (AoA) as well as LED geometric features and identity information to estimate the user's position. Through an experimental setup, the average 3D localization accuracies of 5.58 and 5.44 cm were recorded at heights of 1.45 and 1.75 m, respectively. In [11], the RSSI-based AoA was implemented at different exposure times and a mean positioning accuracy of 7.56 cm was achieved in the experimental demonstration. In [12], a single LED-based Tx detection algorithm was proposed for identifying the edge of the light source based on brightness weakening, Sobel operator, and the least-square error (LSE) fitting, while a smartphone camera was used as the optical Rx. The experimental results revealed that the positioning accuracy is inversely related to the distance as the decoding success rate decreased from 95 to 10% as the distance increased from 1.2 to 2 m. A VLP system using a single LED-based Tx and an IS-based Rx was investigated in [13], where the transmitted signal's RSSI and AoA alongside the fingerprinting-based k-nearest neighbours (k-NN) algorithm were exploited to estimate the position of the Rx. The experimental results demonstrated a 95th percentile accuracy of 1.97 cm.

It is important to note, however, the above-mentioned research studies restrict the use cases of single Tx VLP systems to those that involve only circular LED lighting infrastructures. As a result, a VLP system deploying a single rectangular LED light was proposed in [14], which applied the perspective-n-point (PnP) algorithm on the corners of the rectangular LED to determine the location of the user. Moreover, a corrected geomagnetic angle-visible light positioning (CGA-VLP) algorithm with a square LED as the Tx was proposed in [15], exhibiting a positioning error of less than 10 cm. A localization algorithm considering an unbalanced single LED Tx was studied in [8], obtaining localization accuracies of 2.55, 2.44, and 2.26 m for heights of 2, 2.5, and 3 m, respectively.

In order to overcome the limitation of using multiple LED Tx's, IMU sensors, such as magnetometers or gyroscopes, were implemented on a smart device to estimate the Rx's orientation, i.e., azimuth angles. In [16], a smartphone's IMU sensor was utilized to estimate the user's orientation with the single circular LED Tx. The experimental results showed a positioning accuracy of less than 16 cm. In [17], the smartphone camera and IMU sensor were utilized with pedestrian dead reckoning algorithm for the localization purposes with single LED Tx, in which 2D and 3D positioning accuracies of 2.46 and 13.4 cm

TABLE I – Single LED-based VLP systems

Ref.	Rx specifications	Auxiliary devices	Positioning technique	Testbed	Testbed size (m ³)	Positioning accuracy (cm)	Ambient light noise
[8]	CMOS IS	No	AoA	Experiment	$1.8 \times 1.8 \times 2$	2.55	No
					$1.8 \times 1.8 \times 2.5$	2.44	
					$1.8 \times 1.8 \times 3$	2.26	
[9]	Smartphone camera	No	Circle projection	Experiment	$3 \times 3 \times 1.5$	12.79	No
					$3 \times 3 \times 2$	17.03	
[10]	Smartphone camera	IMU	Planes intersection	Experiment	$2.7 \times 1.8 \times 1.45$	5.58	No
					$2.7 \times 1.8 \times 1.75$	5.44	
[11]	CMOS IS	No	RSSI, AoA	Experiment	$2 \times 2 \times 2.5$	7.56	No
[12]	Smartphone camera	No	Brightness weakening, Sobel operator, and LSE fitting	Experiment	$- \times - \times 1.2$	-	No
[13]	CMOS IS	No	RSSI, AoA, fingerprinting, and k-NN	Experiment	$0.7 \times 0.3 \times 0.2$	1.97	No
[18]	Smartphone camera	Accelerometer, GMF sensor	RSSI, AoA, and Bokeh effect	Experiment	$1 \times 1 \times 2.4$	< 10	No
[14]	Smartphone camera	IMU	AoA, PnP	Hybrid	$- \times - \times 2$	< 5	No
[15]	Smartphone camera	IMU	CGA-VLP	Hybrid	$1.8 \times 1.8 \times 2.6$	< 10	No
[16]	Smartphone camera	IMU	Image processing	Experiment	$1.8 \times 1.8 \times 2$	< 0.16 m	No
[17]	Smartphone camera	IMU	Pedestrian dead reckoning	Experiment	$14 \times 1.6 \times 2.3$	2D: 2.46 3D: 13.4	No
[19]	Multiple PDs	No	Optical power ratio	Simulation	$1 \times 1 \times 1.5$	2.52	No
[20]	Multiple PDs	No	RSSI, Fingerprinting	Simulation/ Experiment	$3 \times 3 \times 3$	9.27	No
[5]	Multiple PDs	No	RSSI, trilateration	Simulation	$10 \times 10 \times 2.4$	-	No
[21]	Rotatable PD	IMU	RSSI, ML	Simulation	$4 \times 4 \times 3.1$	1.74	No
Proposed model	Multiple PDs	No	RSSI, trilateration	Simulation	$6 \times 6 \times 3$	7	Darkroom
						11	1 source
						13	2 sources

were achieved, respectively [17]. RSSI and AoA techniques were implemented in [18], and a particular frame reception technique was proposed as the bokeh effect. The VLP system utilized an Android platform equipped with an accelerometer and geomagnetic field (GMF) sensor for determining the position of the device, where a positioning accuracy of less than 10 cm was shown through the experimental setup. Nevertheless, magnetometers are inherently sensitive to magnetic disturbances in indoor environments, and gyroscope measurements are subject to drift over time. Accordingly, the VLP algorithms that rely on the IMU sensors are susceptible to inaccurate azimuth angle calculations. It is worth mentioning that in IS-based VLP systems, high positioning accuracy can be achieved, up to several centimeters, but at the cost of complex implementation algorithms requiring increased processing time.

B. PD-based Rx

PD-based VLP systems have received considerable attention in the literature due to their high light sensitivity and relatively low cost [1]. As part of PD-based VLP systems, a variety of methods are available for measuring the position of an object, including time difference of arrival (TDOA), time of arrival (TOA), AoA, and RSSI [7]. Both the TDOA and TOA techniques for indoor localization usually require precise time synchronization between the communication devices, whereas the AoA scheme needs sensor arrays to determine the angle of arrival of the incident signal, which limits the system's efficiency. The RSSI specifies the power level of the received signal, which is directly related to the current gain of the PD and can be obtained without implementing an auxiliary device. Thus, it is widely utilized in VLP systems. In the case of single

LED-based VLP systems implementing RSSI, the multiple Rx's in the form of one horizontal and three tilted PDs have been used in [19] and [20]. In [19], the optical power of multiple PDs was utilized to estimate the user's location with a mean 3D localization accuracy of 2.52 cm, whereas in [20], RSSI with fingerprinting was used to achieve a positioning accuracy of 9.27 cm. The RSSI-based reverse trilateration scheme was proposed in [5], in which three PDs arranged in a circular geometry were deployed for positioning purposes, and the simulation results showed that a high signal-to-noise ratio (SNR) is required to achieve low positioning errors. In [21], a rotatable PD alongside an RSSI-based machine learning (ML) model was exploited to obtain a positioning accuracy of 1.74 cm. It is worth mentioning that the above-mentioned PD-based VLP techniques were developed for the downlink scenario without the consideration of ambient light in the system.

III. PROPOSED SINGLE LED-BASED VLP SYSTEM

The study focuses on investigating a VLP system that employs a single LED Tx. It is assumed that the LED Tx on the ceiling of the given cube-shaped indoor environment is oriented downward, whereas the plane of the optical Rx at the user, which is considered to be located at the floor level, is directed upward, see Fig. 1. The Tx projection on the 2D xy plane of the indoor environment is considered to be known and represented as $\mathbf{X}_{Tx} = (x_{Tx}, y_{Tx})'$. An optical Rx with four identical PDs arranged in a symmetrical pattern following a rectangular geometry is considered in this work. In this case, the center of the Rx, i.e., the midpoint of the diagonal of the Rx's rectangular surface with respect to the xy plane of the given indoor environment is represented as $\mathbf{X}_{Rx} = (x_{Rx}, y_{Rx})'$, and it corresponds to the ground

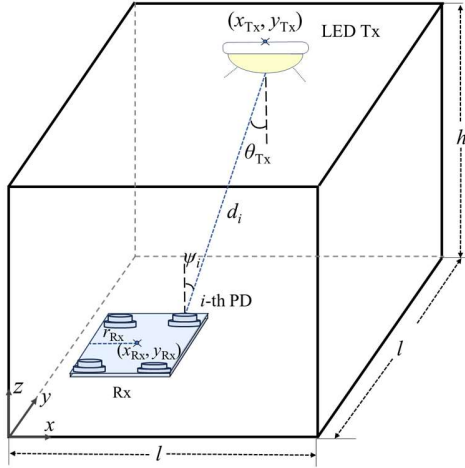


Fig. 1. Proposed single LED-based VLP Model.

truth for the position estimation of the user. It is considered, without the loss of generality, that each PD is equidistant with a radius, r_{Rx} , from the center of the Rx rectangular plane. The 2D coordinates of the i -th PD with respect to r_{Rx} , given that $i = \{1, 2, 3, 4\}$, can then be represented as $\mathbf{X}_i = (x_i, y_i)'$, where $\mathbf{X}_1 = (-r_{Rx}, r_{Rx})'$, $\mathbf{X}_2 = (r_{Rx}, r_{Rx})'$, $\mathbf{X}_3 = (-r_{Rx}, -r_{Rx})'$, and $\mathbf{X}_4 = (r_{Rx}, -r_{Rx})'$. It should be noted that bold characters denote a row vector, and the vector transposition is indicated by the superscript $\{.\}'$.

A. Channel Model

In order to estimate the user location, the LED-based Tx is designed to constantly transmit unique information packets via an LOS optical channel, which carries information on $\mathbf{X}_{Tx}$. It is noted that the spatial diversity of the PDs at the user's Rx allows the parallel reception of the data, in which the received signal at a given PD is defined based on the inverse-square law associated with a free-space Lambertian channel model, given as [22]:

$$f_i(t) = \gamma_i s(t - \tau_i) + n_i(t), \quad (1)$$

$$\gamma_i = A_i R_i \cos(\theta_{Tx})^m \cos(\psi_i) T(\psi_i) g(\psi_i) \times \frac{(m+1)}{2\pi d_i^2}, \quad (2)$$

where the transmitted optical signal is denoted by $s(t)$, while τ_i and $n_i(t)$ are the time of arrival and noise at the i -th PD, respectively. The i -th PD's active physical area and responsivity are designated as A_i and R_i , respectively. The angle of irradiance of the LED Tx is denoted as θ_{Tx} . Furthermore, ψ_i is the angle of incidence at the i -th PD. $T(\psi_i)$ is the signal transmission of the optical filter, and $g(\psi_i)$ signifies the gain of an optical concentrator. It is also important to note that ψ_i should be within the FOV of the respective PD, and there are no optical lenses implemented in the proposed system. Thus, the values of $T(\psi_i)$ and $g(\psi_i)$ are set to unity. The distance from the LED Tx to the i -th PD at the Rx of the user is represented by d_i , and m corresponds to the Lambertian order of the LED Tx, which is given as [23]:

$$m = -\frac{\ln(2)}{\ln(\cos \phi_{1/2-Tx})}, \quad (3)$$

where $\phi_{1/2-Tx}$ is the semi-angle at half-power of the LED Tx.

B. Noise Model

The noise term $n_i(t)$ in (1) represents the incorporation of various noises affecting the received optical signal at the i -th PD of the user's Rx. In addition to the usual thermal noise, several other noises occur in the PD as it picks up optical signals, including the inherent shot noise, the noise generated by the dark current, the noise originating from electromagnetic background radiation, and the noise induced by the ambient light sources in the indoor environment. In line with standard practice, it is assumed that these noises follow Gaussian distribution, and the variances for the respective noises in the proposed system are given by [22], [23]:

$$\sigma_{\text{thermal-}i}^2 = \frac{8\pi \kappa T_k}{G_i} C_i A_i I_2 B_i^2 + \frac{16\pi^2 \kappa T_k \Gamma}{g_m} C_i^2 A_i^2 I_3 B_i^3, \quad (4)$$

$$\sigma_{\text{shot-}i}^2 = 2q B_i \gamma_i P_{Tx}, \quad (5)$$

where q is the electronic charge, κ is Boltzmann's constant, and T_k depicts the absolute temperature of the given indoor environment. G_i , B_i , and C_i indicate the open-loop voltage gain, bandwidth, and the fixed capacitance per unit area of the i -th PD, respectively. Γ and g_m correspond to the field effect transistor's (FET) channel noise factor and transconductance, respectively. I_2 and I_3 are the characteristic constants representing the noise bandwidth factor, and P_{Tx} is the optical power of the transmitted signal. It can be observed from (5) that the variance of the shot noise is proportional to the strength of the received optical signal at the i -th PD. As to the dark current and the background radiation noises, the variances are given as [22]:

$$\sigma_{\text{dark-}i}^2 = 2q I_{\text{dark}} B_i, \quad (6)$$

$$\sigma_{\text{back-}i}^2 = 2q A_i B_i R_i \Delta \lambda v_{\text{back}}, \quad (7)$$

where I_{dark} is the PD dark current, $\Delta \lambda$ is the bandwidth of the optical filter, and v_{back} is the background spectral irradiance. Moreover, various natural and artificial sources such as sunlight and adjacent illumination fixtures, contribute to the ambient light causing noise in the free space optical channel of the VLC system. The proposed VLP system is implemented in an enclosed indoor environment hence, artificial sources are the dominant factor of ambient light noise in this study. The variance of the unmodulated ambient light source causing noise in the system is given by [24]:

$$\sigma_{\text{amb-}i}^2 = 2q B_i \gamma_{\text{amb-}i} P_{\text{amb}}, \quad (8)$$

where $\gamma_{\text{amb-}i}$ represents the optical channel model between the ambient light source and the i -th PD, whereas P_{amb} denotes the transmitted optical power of the source.

C. Localization Model

When the received optical signal indicating the RSSI and thus, the d_i between each pair of LED-Tx and the i -th PD at the Rx is measured, the proposed localization model utilizes the trilateration algorithm to ultimately estimate the user position in the indoor environment, given as [6]:

$$\Delta r_i^2 + h^2 = d_i^2, \quad (9)$$

$$\Delta r_i^2 = (x_{Tx} - (\hat{x}_{Rx} + x_i))^2 - (y_{Tx} - (\hat{y}_{Rx} + y_i))^2, \quad (10)$$

where $(\hat{x}_{Rx}, \hat{y}_{Rx})$ indicate the estimated user coordinates, i.e., the center of the rectangular plane at the Rx surface. Equations 9 and 10 follow the non-linear least square estimation scheme, which can also be represented as a closed-form matrix solution given by [5], [6]:

$$\hat{\mathbf{X}}_{Rx} = (\mathbf{a}'\mathbf{a})^{-1}\mathbf{a}'\mathbf{\beta}, \quad (11)$$

where,

$$\mathbf{a} = \begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \\ \vdots & \vdots \\ x_i - x_1 & y_i - y_1 \end{bmatrix}, \quad (12)$$

$$\mathbf{\beta} = 0.5 \times \begin{bmatrix} (\Delta r_1^2 - \Delta r_2^2) + (a_2^2 + b_2^2) - (a_1^2 + b_1^2) \\ (\Delta r_1^2 - \Delta r_3^2) + (a_3^2 + b_3^2) - (a_1^2 + b_1^2) \\ \vdots \\ (\Delta r_1^2 - \Delta r_i^2) + (a_i^2 + b_i^2) - (a_1^2 + b_1^2) \end{bmatrix}, \quad (13)$$

where $a_i = x_{Tx} - x_i$ and $b_i = y_{Tx} - y_i$.

IV. PERFORMANCE EVALUATION

The performance of the proposed LOS single LED-based localization model is evaluated using numerical simulations, on the MATLAB[®] simulation platform, following the channel and positioning model described in Section III. Table II enlists the default system parameters for the simulation environment. It can be observed by implementing the parameters in Table II that the dark current and background radiation noise factors are insignificant, and it is also worth noting that the VLC systems primarily operate in the visible light spectrum, and most of the background radiation noise from non-visible light frequencies might not have a significant impact on the system's performance. On that note, the system analysis considers the thermal, shot, and potential ambient light noise factors influencing the positioning performance.

The simulation of the proposed system is implemented for the cases of (a) ambient-noise-free region (i.e., dark room), in which the single LED-based Tx is transmitting the optical signal to the user's Rx, (b) single ambient light source, and (c) two ambient light sources, where the identical light sources are

Parameter	Symbol	Value
Room size	(l, l, h)	$(6 \times 6 \times 3) \text{ m}^3$
LED Tx coordinates	(x_{Tx}, y_{Tx})	$(3, 3) \text{ m}$
LED-1 ambient source coordinates	(x_{1-a}, y_{1-a})	$(1, 3) \text{ m}$
LED-2 ambient source coordinates	(x_{2-a}, y_{2-a})	$(5, 3) \text{ m}$
LED Tx optical power	P_{Tx}	1 W
LED Tx semi-angle at half power	$\phi_{1/2-Tx}$	60°
Rx radius	r_{Rx}	0.1 m
PD field-of-view	FOV	60°
PD active physical area	A_i	10^{-4} m^2
PD bandwidth	B_i	10 MHz
PD capacitance	C_i	1.69 pF
PD responsivity	R_i	0.5 A/W
Open-loop voltage gain	G_i	1
PD's dark current	I_{dark}	5 pA
Background spectral irradiance	ν_{back}	$5.8 \times 10^{-2} \text{ W/m}^2/\text{nm}$
FET transconductance	g_m	$15 \times 10^{-3} \text{ S}$
FET channel noise factor	Γ	2×10^{-8}
Room temperature	T_k	298.15 K
Characteristic constant	I_2	0.562
Characteristic constant	I_3	0.0868

present in the given indoor environment for the illumination purpose besides the optical Tx. It is considered for aesthetic purposes that the light fixtures in the room are located in line with each other, and the xy coordinates of the optical sources on the ceiling are known at the time of installation.

Fig. 2. (a), (b) and (c) depict the spatial distribution of the estimated position with respect to the ground truth, for a grid of possible Rx locations in the given indoor environment considering the cases of a dark room, one ambient light source, and two ambient light sources, respectively. Fig. 2. shows that the existence of ambient light in the system affects the positioning accuracy of the system. The efficiency of the system is analyzed in terms of the key parameter of root mean square positioning error, given as:

$$\delta_p = \sqrt{(x_{Rx} - \hat{x}_{Rx})^2 + (y_{Rx} - \hat{y}_{Rx})^2}. \quad (14)$$

Fig. 3. demonstrates the empirical distribution, i.e., cumulative distribution function (CDF), of δ_p for the given cases of ambient light in the room. It is observed that a 90th percentile accuracy of 7, 11, and 13 cm is achieved for the given indoor environment in the case of a dark room, one ambient light, and two ambient light sources, respectively. Hence, a reduction of

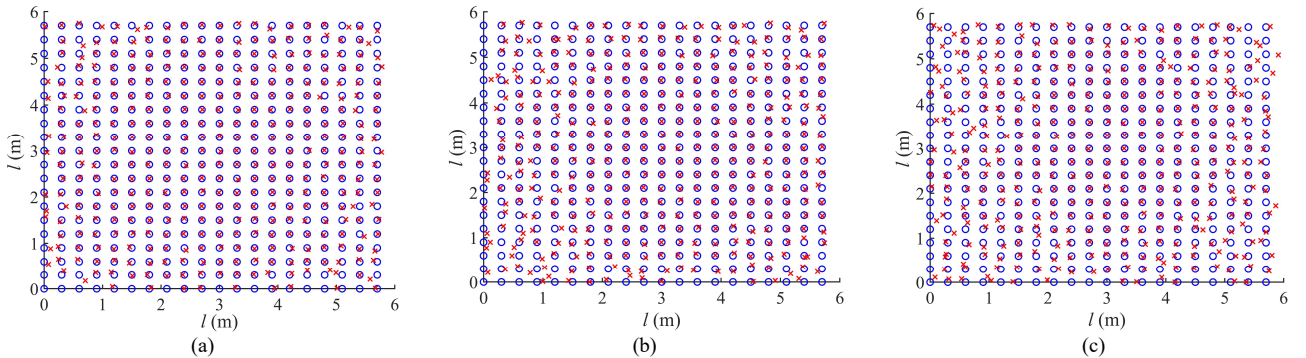


Fig. 2. Spatial distribution of the estimated positioning error $(\hat{x}_{Rx}, \hat{y}_{Rx})$, red marks, with respect to the ground truth (x_{Rx}, y_{Rx}) , blue circles, in different ambient light conditions: (a) ambient-noise-free region (i.e., dark room); (b) single ambient light source at $(1, 3)$; and (c) two ambient light sources at $(1, 3)$ and $(5, 3)$.

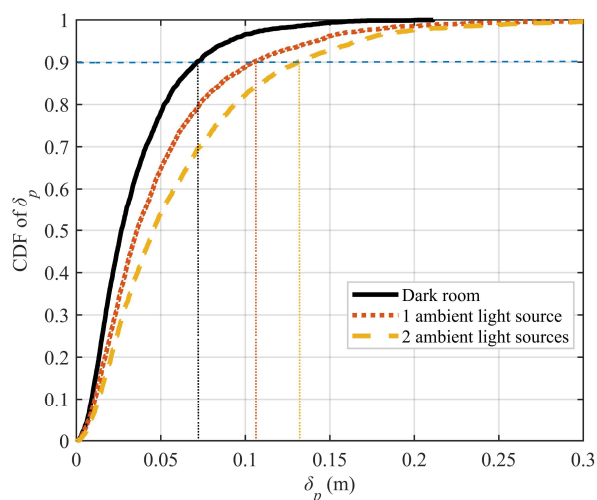


Fig. 3. Positioning precision as a function of δ_p for the dark room and ambient light cases.

approximately 46% in δ_p can be observed without the presence of ambient light in the system.

V. CONCLUSION

In this work, the positioning performance of a single LED-based indoor VLP system was investigated in the presence of ambient light. The paper discussed the key research progress in the field and a distinctive indoor positioning algorithm was implemented where the single LED-based optical Tx was employed to send the free-space light signal to the user's optical Rx, with an unknown location, which was equipped with four PDs acting as optical sensors. Once the user evaluated the RSSI, the free-space optical channel model was used to approximate the distance between the LED Tx and the i -th PD. Consequently, the estimated distance was utilized by the positioning algorithm, i.e., trilateration, to locate the user in the given indoor environment. Through simulation results, a 46% reduction in the positioning error was observed for the case of an ambient-noise-free region (i.e., a dark room) in comparison to that of a well-illuminated area.

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REFERENCES

- [1] Timothy H. Kindervatter and Fernando L. Teixeira, "Overview of the Global Positioning System," in *Tropospheric and Ionospheric Effects on Global Navigation Satellite Systems*, IEEE, 2022, pp. 1–42. doi: 10.1002/9781119863069.ch1.
- [2] F. Alam, N. Faulkner, and B. Parr, "Device-Free Localization: A Review of Non-RF Techniques for Unobtrusive Indoor Positioning," *IEEE Internet Things J.*, vol. 8, no. 6, pp. 4228–4249, Mar. 2021, doi: 10.1109/JIOT.2020.3030174.
- [3] N. Chaudhary, L. N. Alves, and Z. Ghassemloooy, "Current Trends on Visible Light Positioning Techniques," in *2019 2nd West Asian Colloquium on Optical Wireless Communications (WACOWC)*, Apr. 2019, pp. 100–105. doi: 10.1109/WACOWC.2019.8770211.
- [4] Y. Zhuang *et al.*, "A Survey of Positioning Systems Using Visible LED Lights," *IEEE Commun. Surv. Tutor.*, vol. 20, no. 3, pp. 1963–1988, thirdquarter 2018, doi: 10.1109/COMST.2018.2806558.
- [5] N. Chaudhary, L. N. Alves, and Z. Ghassemloooy, "Feasibility Study of Reverse Trilateration Strategy with a Single Tx for VLP," in *2019 2nd West Asian Colloquium on Optical Wireless Communications (WACOWC)*, Apr. 2019, pp. 121–126. doi: 10.1109/WACOWC.2019.8770213.
- [6] N. Chaudhary, O. I. Younus, L. N. Alves, Z. Ghassemloooy, S. Zvanovec, and H. Le-Minh, "An Indoor Visible Light Positioning System Using Tilted LEDs with High Accuracy," *Sensors*, vol. 21, no. 3, 2021, doi: 10.3390/s21030920.
- [7] M. Maheepala, A. Z. Kouzani, and M. A. Joordens, "Light-Based Indoor Positioning Systems: A Review," *IEEE Sens. J.*, vol. 20, no. 8, pp. 3971–3995, Apr. 2020, doi: 10.1109/JSEN.2020.2964380.
- [8] H. Li *et al.*, "A Fast and High-Accuracy Real-Time Visible Light Positioning System Based on Single LED Lamp With a Beacon," *IEEE Photonics J.*, vol. 12, no. 6, pp. 1–12, Dec. 2020, doi: 10.1109/JPHOT.2020.3032448.
- [9] R. Zhang, W. -D. Zhong, Q. Kemao, and S. Zhang, "A Single LED Positioning System Based on Circle Projection," *IEEE Photonics J.*, vol. 9, no. 4, pp. 1–9, Aug. 2017, doi: 10.1109/JPHOT.2017.2722474.
- [10] H. Cheng, C. Xiao, Y. Ji, J. Ni, and T. Wang, "A Single LED Visible Light Positioning System Based on Geometric Features and CMOS Camera," *IEEE Photonics Technol. Lett.*, vol. 32, no. 17, pp. 1097–1100, Sep. 2020, doi: 10.1109/LPT.2020.3012476.
- [11] O. I. Younus *et al.*, "A unilateral 3D indoor positioning system employing optical camera communications," *IET Optoelectron.*, vol. n/a, no. n/a, doi: https://doi.org/10.1049/ote.2.12094.
- [12] Y. Ji *et al.*, "A single LED lamp positioning system based on CMOS camera and visible light communication," *Opt. Commun.*, vol. 443, pp. 48–54, 2019, doi: https://doi.org/10.1016/j.optcom.2019.03.002.
- [13] Z. Zheng, L. Liu, C. Zhao, and W. Hu, "High accuracy indoor positioning scheme using single LED and camera," *Electron. Lett.*, vol. 54, no. 4, pp. 227–229, 2018, doi: https://doi.org/10.1049/el.2017.3012.
- [14] B. Hussain, Y. Wang, R. Chen, and C. P. Yue, "Camera Pose Estimation Using a VLC-Modulated Single Rectangular LED for Indoor Positioning," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022, doi: 10.1109/TIM.2022.3212980.
- [15] C. Yang, S. Wen, D. Yuan, J. Chen, J. Huang, and W. Guan, "CGA-VLP: High Accuracy Visible Light Positioning Algorithm Using Single Square LED with Geomagnetic Angle Correction," *Photonics*, vol. 9, no. 9, 2022, doi: 10.3390/photonics9090653.
- [16] J. Hao, J. Chen, and R. Wang, "Visible Light Positioning Using A Single LED Luminaire," *IEEE Photonics J.*, vol. 11, no. 5, pp. 1–13, Oct. 2019, doi: 10.1109/JPHOT.2019.2930209.
- [17] H. Huang, B. Lin, L. Feng, and H. Lv, "Hybrid indoor localization scheme with image sensor-based visible light positioning and pedestrian dead reckoning," *Appl. Opt.*, vol. 58, no. 12, pp. 3214–3221, Apr. 2019, doi: 10.1364/AO.58.003214.
- [18] J. -W. Lee, S. -J. Kim, and S. -K. Han, "3D Visible Light Indoor Positioning by Bokeh Based Optical Intensity Measurement in Smartphone Camera," *IEEE Access*, vol. 7, pp. 91399–91406, 2019, doi: 10.1109/ACCESS.2019.2927356.
- [19] L. Qin, B. Niu, B. Li, and Y. Du, "Indoor visible light high precision three-dimensional positioning algorithm based on single LED lamp," *Optik*, vol. 207, p. 163786, Apr. 2020, doi: 10.1016/j.ijleo.2019.163786.
- [20] L. Qin, B. Niu, B.-S. Li, X.-L. Hu, and Y.-X. Du, "High precision indoor positioning algorithm of single LED lamp based on A-Bayes," *Optik*, vol. 241, p. 167190, Sep. 2021, doi: 10.1016/j.ijleo.2021.167190.
- [21] R. Liu, Z. Liang, K. Yang, and W. Li, "Machine Learning Based Visible Light Indoor Positioning With Single-LED and Single Rotatable Photo Detector," *IEEE Photonics J.*, vol. 14, no. 3, pp. 1–11, Jun. 2022, doi: 10.1109/JPHOT.2022.3163415.
- [22] X. Liu, D. Zou, N. Huang, and S. Zhang, "A Comprehensive Accuracy Analysis of Visible Light Positioning under Shot Noise," in *2020 IEEE/CIC International Conference on Communications in China (ICCC Workshops)*, Aug. 2020, pp. 167–172. doi: 10.1109/ICCCWorkshops49972.2020.9209923.
- [23] Z. Ghassemloooy, W. Popoola, and S. Rajbhandari, *Optical Wireless Communications: System and Channel Modelling with MATLAB*, Second Edition. CRC Press, Inc., 2019.
- [24] S. Kadirvelu and V. B., "Review On Impact Of Ambient Light Noise Sources And Applications In Optical Wireless Communication Using LED," *Int. J. Appl. Eng. Res.*, vol. 10, pp. 31115–31130, Jul. 2015.